Highlights

Bridging the Digital Divide: Asymmetric Broadband Price Elasticity in the European Union and Eastern Partnership Countries

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- Eastern Partnership countries exhibit significantly stronger price elasticity (-0.608^{***}) than EU countries (-0.054, not significant)
- EaP countries are 11.3 times more price-responsive, with interaction term highly significant ($p<0.001$)
- Results robust across 8 model specifications with alternative controls and time periods
- COVID-19 pandemic had asymmetric regional impact: EaP experienced 3.7× stronger adoption boost
- Policy recommendation: Target price subsidies to EaP where effective; focus EU policy on quality and digital skills

Bridging the Digital Divide: Asymmetric Broadband Price Elasticity in the European Union and Eastern Partnership Countries

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Abstract

Does broadband price sensitivity differ systematically between developed and developing European regions? Using a two-way fixed effects panel model with region-price interaction terms, we analyze 33 countries (27 EU member states and 6 Eastern Partnership nations) over 2010-2019. We find striking asymmetry: Eastern Partnership (EaP) countries exhibit a price elasticity of -0.608 ($p<0.001$), compared to -0.054 ($p=0.17$, statistically insignificant) in EU countries—an 11.3-fold difference. A 10% broadband price reduction would increase internet adoption by 6.1% in EaP countries but only 0.5% in the EU. These findings persist across eight model specifications. Extended analysis through 2023 reveals that EaP countries experienced 3.7 times stronger pandemic-induced adoption acceleration. Our results demonstrate that price-based policy interventions exhibit sharply diminishing returns at higher development levels, with critical implications for targeted digital infrastructure strategies in the EU Digital Decade agenda and EaP connectivity programs.

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1. Introduction

The persistent digital divide between developed and developing European regions remains one of the most pressing policy challenges in telecommunications, despite decades of infrastructure investment and regulatory reform [2]. While internet penetration exceeds 85% in most Western European nations, Eastern Partnership (EaP) countries—Armenia, Azerbaijan, Belarus, Georgia, Moldova, and Ukraine—continue to lag substantially, with average adoption rates around 65% during the pre-pandemic period. Understanding the fundamental drivers of these disparities is essential for designing effective interventions to achieve universal connectivity goals outlined in the EU Digital Decade agenda and EaP digital transformation initiatives.

Price has long been recognized as a critical determinant of broadband adoption, yet existing empirical evidence reveals considerable heterogeneity in estimated price elasticities across countries, regions, and development levels [3, 2]. Studies from Latin America report elasticities ranging from -1.79 to -2.20, suggesting highly price-sensitive demand [4], while OECD countries typically exhibit values between -0.36 and -0.59, indicating relatively inelastic demand [1]. This substantial variation raises a fundamental question: does the relationship between broadband prices and adoption differ systematically based on economic development level and market maturity?

Despite extensive research on broadband demand estimation, a critical gap remains in the literature. Prior studies typically analyze individual coun-

tries or homogeneous regional groupings, implicitly assuming that price elasticity is constant within their sample [5, 3]. No study has explicitly tested for regional heterogeneity in price responsiveness by employing interaction models that formally assess whether elasticity differs between developed and developing European regions. This methodological limitation prevents researchers and policymakers from quantifying the magnitude of heterogeneity and testing whether observed differences are statistically significant or merely sampling variation.

This paper fills this gap by directly comparing broadband price elasticity between EU and EaP countries using a unified panel framework with region-price interaction terms. Our research questions are: (1) Does broadband price elasticity differ significantly between EU and EaP countries? (2) What is the magnitude of this difference, and does it persist across alternative specifications, control variables, and time periods? (3) How did the COVID-19 pandemic—which accelerated digital transformation globally—differentially affect broadband adoption across these regions?

We make several contributions to the telecommunications policy literature. First, we document striking regional heterogeneity in broadband price elasticity: EaP countries exhibit an average elasticity of -0.608 ($p<0.001$), compared to -0.054 ($p=0.17$, statistically insignificant) in EU countries—an 11.3-fold difference. This finding emerges from a two-way fixed effects panel model with country and year fixed effects, controlling for GDP per capita, R&D expenditure, and digital infrastructure quality, with standard errors clustered at the country level to account for serial correlation. The interaction term between price and the EaP region dummy is highly statistically

significant ($p<0.001$), confirming that the elasticity differential is not due to chance.

Second, we demonstrate that this result is robust across eight alternative model specifications, including separate regional regressions with different control variable combinations, full-period analysis extending through 2023, and models incorporating COVID-19 controls and COVID×Region interactions. In every specification, EaP countries exhibit substantially stronger price responsiveness than EU countries, confirming that our finding is not driven by outliers, particular control choices, or sample period selection.

Third, we provide novel evidence on differential pandemic impacts. Extended analysis through 2023 with COVID×Region interaction terms reveals that EaP countries experienced 3.7 times stronger adoption acceleration during the pandemic (+35.7% vs. +9.6% for EU), suggesting they had larger pools of potential adopters held back by pre-pandemic barriers that COVID-19 necessity temporarily overcame.

The policy implications are profound and actionable. A 10% reduction in broadband prices would increase internet adoption by 6.1% in EaP countries (highly effective) but only 0.5% in the EU (negligible and statistically insignificant). This stark asymmetry demonstrates that one-size-fits-all regional policies are inefficient: price subsidies and affordability programs should be targeted to EaP countries where they generate substantial returns, while EU policy should pivot toward quality improvements, speed upgrades, and digital skills development where price has minimal marginal impact.

The remainder of this paper proceeds as follows. Section 2 reviews the literature on broadband demand estimation, price elasticity heterogeneity,

and the European digital divide. Section 3 describes our data sources, variable construction, and descriptive statistics. Section 4 presents our empirical strategy, including the two-way fixed effects model with region-price interaction and robustness specifications. Section 5 reports our main findings and eight robustness checks. Section 6 interprets the economic mechanisms underlying the 11.3-fold elasticity differential and compares our results to prior literature. Section 7 discusses policy implications for EU and EaP digital strategies. Section 8 concludes with limitations and directions for future research.

2. Literature Review

2.1. Broadband Demand Estimation

The economics literature on broadband demand estimation has evolved considerably over the past two decades, moving from simple cross-sectional analyses to sophisticated panel data methods that address endogeneity concerns and exploit within-country variation over time [5]. Early studies focused primarily on OECD countries, where data availability was superior and regulatory frameworks more transparent [1]. These studies consistently found that price, income, and infrastructure quality were significant determinants of broadband adoption, though estimated coefficients varied substantially across specifications and countries.

Cardona et al. [1] provide a comprehensive analysis of broadband demand across OECD countries, estimating price elasticities ranging from -0.36 to -0.59 depending on specification and technology type (DSL vs. cable). Their fixed effects panel approach controls for unobserved country-specific factors,

representing a methodological advancement over earlier cross-sectional studies that potentially suffered from omitted variable bias. However, their analysis assumes elasticity homogeneity across all OECD members, precluding tests for systematic differences between higher and lower-income countries within the sample.

More recent work has extended demand estimation to emerging markets and developing regions. Galperin and Mariscal [2] analyze Latin American countries and report substantially higher price elasticities than observed in OECD studies, with estimates ranging from -0.36 to -0.48 for fixed broadband. They attribute this difference to lower income levels, earlier stages in the adoption S-curve, and greater budget constraints binding on household technology purchases. Similarly, Macedo et al. [4] estimate Brazilian municipal-level elasticities between -1.79 and -2.16, confirming highly price-sensitive demand in developing contexts.

2.2. Price Elasticity Heterogeneity

The substantial variation in estimated price elasticities across studies and contexts raises important questions about the sources of heterogeneity. Several theoretical mechanisms predict that elasticity should vary systematically with economic development level [3]. First, income effects suggest that broadband price represents a larger share of disposable income in lower-income countries, amplifying price sensitivity as budget constraints bind more tightly. Second, the S-curve diffusion model predicts that elasticity is highest at intermediate adoption levels (around 50%) and lower at both very low and very high penetration, as marginal adopters become increasingly dissimilar from the majority population at the extremes.

Third, market structure and competitive intensity may influence price responsiveness. Markets with limited competition and high markups should exhibit stronger consumer reactions to price changes, as reductions move closer to competitive equilibrium prices. Fourth, the quality-adjusted nature of broadband services complicates elasticity interpretation: if price reductions are accompanied by quality improvements (faster speeds, better reliability), observed adoption increases may partly reflect quality effects rather than pure price effects [3].

Despite these theoretical predictions, most empirical studies do not explicitly test for heterogeneity. By analyzing single countries or assuming coefficient homogeneity within regional groupings, they cannot quantify elasticity differences or assess statistical significance. The few studies that do examine heterogeneity typically split samples post-hoc rather than employing interaction models that formally test for differential effects [5].

2.3. Digital Divide in Europe

The European digital divide manifests along multiple dimensions: geographic (urban vs. rural), socioeconomic (income, education), demographic (age, gender), and regional (Western vs. Eastern Europe). The European Commission's Digital Economy and Society Index (DESI) documents persistent gaps in connectivity, digital skills, internet use, and digitalization of businesses and public services across EU member states. While Nordic countries consistently rank highest, Southern and Eastern European nations lag on multiple indicators.

The Eastern Partnership—a joint initiative between the EU and six Eastern European countries (Armenia, Azerbaijan, Belarus, Georgia, Moldova,

Ukraine)—explicitly aims to strengthen political and economic ties while supporting digital transformation. The EU4Digital program focuses on harmonizing digital markets, improving regulatory frameworks, expanding broadband access, and enhancing cybersecurity. However, EaP countries face structural challenges including lower per capita incomes (3-4 times below EU averages), weaker institutional capacity, less developed digital infrastructure, and more volatile political environments.

Understanding whether and how broadband demand differs between EU and EaP countries is critical for designing effective policies under these initiatives. If price elasticity is substantially higher in EaP countries, affordability programs and price subsidies would generate larger adoption returns than in saturated EU markets. Conversely, if elasticities are similar, other barriers (infrastructure gaps, digital literacy, content availability) may be more binding constraints requiring different policy tools.

2.4. COVID-19 and Digital Transformation

The COVID-19 pandemic dramatically accelerated digital transformation as lockdowns, remote work, and online education became necessities. Internet traffic surged globally, videoconferencing adoption skyrocketed, and e-commerce penetration increased years ahead of pre-pandemic projections. However, the extent to which this digital acceleration differentially affected developed versus developing regions remains empirically underexplored.

Some evidence suggests the pandemic widened digital divides by exacerbating pre-existing inequalities: households without broadband access faced severe disadvantages in remote education and work, while lower-income countries struggled with infrastructure capacity constraints during traffic surges.

Other research points to narrowing gaps as necessity drove rapid adoption among previously reluctant or constrained populations. Distinguishing these competing hypotheses requires explicit modeling of COVID impacts with region-specific coefficients.

2.5. Research Gap and Contribution

Despite this extensive literature, no study has: (1) directly compared EU and EaP countries using formal interaction models that explicitly test for elasticity heterogeneity; (2) quantified the magnitude of elasticity differences with rigorous statistical significance tests; (3) examined multiple robustness specifications to confirm findings are not driven by particular modeling choices; or (4) analyzed differential COVID-19 impacts on broadband adoption across these regions.

We address these gaps by employing a unified two-way fixed effects panel framework with price×region interaction terms, comprehensive robustness checks across eight alternative specifications, and extended analysis through the pandemic period with COVID×region interactions. Our approach allows us to formally test whether EaP countries are significantly more price-elastic than EU countries, quantify the magnitude of this difference, and assess whether it persists across alternative controls, samples, and time periods.

3. Data and Variables

3.1. Sample Coverage

Our sample comprises 33 European countries observed annually from 2010 to 2023, yielding a maximum of 462 country-year observations. The sample

includes 27 European Union member states and 6 Eastern Partnership countries (Armenia [ARM], Azerbaijan [AZE], Belarus [BLR], Georgia [GEO], Moldova [MDA], and Ukraine [UKR]). We focus our main analysis on the pre-COVID period (2010-2019) to avoid pandemic-related distortions, yielding 330 potential observations. After removing observations with missing data on key variables, our baseline estimation sample contains 319 country-year observations.

For robustness checks, we extend the analysis through 2023 to examine the full post-pandemic period, yielding 451 observations after accounting for missing data. This extended sample allows us to test whether our findings persist when including COVID-19 years and to explicitly model differential pandemic impacts using COVID×region interaction terms.

3.2. Dependent Variable

Our dependent variable is the natural logarithm of the percentage of the population using the internet (`log_internet_users_pct`). This measure captures broadband adoption at the extensive margin—whether individuals have internet access—rather than intensive margin usage patterns. We obtain internet user data from the International Telecommunication Union (ITU) World Telecommunication/ICT Indicators Database, which provides internationally comparable statistics based on standardized definitions and collection methodologies.

The log transformation serves three purposes. First, it yields coefficients interpretable as elasticities, allowing us to directly assess percentage changes in adoption associated with percentage price changes. Second, it reduces heteroskedasticity by compressing the scale of the dependent variable, which

ranges from below 20% in some EaP countries in early years to above 95% in Nordic countries in recent years. Third, it better approximates the S-curve pattern of technology diffusion, which is inherently nonlinear but approximately log-linear over intermediate adoption ranges.

3.3. Key Independent Variables

Broadband Price. Our primary independent variable is the natural logarithm of PPP-adjusted fixed broadband subscription prices (`log_fixed_broad_price`), measured in current international dollars. Price data come from the ITU database, which collects information on monthly subscription costs for entry-level fixed broadband packages meeting minimum speed thresholds (typically 256 kbit/s download). PPP adjustment accounts for cross-country differences in purchasing power, ensuring that nominal price differences reflect real affordability variations rather than exchange rate fluctuations or general price level differences.

Region Indicator. We construct a binary variable `eap_region` equal to 1 for Eastern Partnership countries (Armenia, Azerbaijan, Belarus, Georgia, Moldova, Ukraine) and 0 for EU member states. This indicator allows us to estimate differential intercepts and, crucially, differential price slopes through the interaction term.

Price×Region Interaction. The interaction between `log_fixed_broad_price` and `eap_region` is our coefficient of primary interest. A statistically significant negative coefficient on this term indicates that EaP countries exhibit stronger (more negative) price elasticity than EU countries, confirming regional heterogeneity in price responsiveness.

3.4. Control Variables

Following the broadband demand literature [1, 2], we include three time-varying control variables that capture economic development, innovation capacity, and digital infrastructure quality.

GDP per capita (PPP-adjusted, log). Higher income levels are associated with greater broadband adoption through multiple channels: increased affordability, higher valuation of digital services, complementarities with skilled employment, and general technology readiness. We use PPP-adjusted GDP per capita in constant international dollars from the World Bank World Development Indicators (WDI), log-transformed to account for nonlinear income effects.

R&D expenditure (% of GDP). Research and development intensity proxies for a country's innovation capacity, technological sophistication, and human capital endowment. Countries with stronger R&D sectors typically exhibit higher demand for broadband due to complementarities between advanced digital infrastructure and knowledge-intensive economic activities. We obtain R&D expenditure as a percentage of GDP from the World Bank WDI.

Secure internet servers (per million people, log). The number of secure internet servers per capita proxies for digital infrastructure quality, e-commerce readiness, and the availability of online services that create demand for broadband access. Countries with more extensive server infrastructure offer richer online content and services, increasing the value proposition of internet adoption. We use World Bank WDI data, log-transformed to reduce skewness.

3.5. COVID-19 Variable

For extended-period robustness checks (2010-2023), we construct a binary COVID-19 indicator equal to 1 for years 2020-2023 and 0 otherwise. This allows us to control for the pandemic's aggregate impact on broadband adoption. We also construct a COVID×Region interaction to test whether pandemic effects differed between EU and EaP countries.

3.6. Descriptive Statistics

Table 1 presents summary statistics for the main analysis sample (2010-2019, N=319). Several patterns emerge. First, mean internet adoption is substantially higher in EU countries (73.8%) than EaP countries (51.2%), confirming a persistent digital divide. Second, despite lower adoption, EaP countries have lower mean broadband prices ($11.85 vs. $27.31 for EU), though this difference partly reflects PPP adjustment for lower general price levels.

Third, EaP countries exhibit much lower GDP per capita ($8,420 vs. $34,200 for EU), confirming that they are at earlier stages of economic development. Fourth, R&D intensity is substantially lower in EaP countries (0.48% of GDP) than EU (1.89%), reflecting weaker innovation ecosystems. Fifth, secure internet server density is dramatically lower in EaP (119 per million) versus EU (1,247 per million), indicating less developed digital infrastructure and online service ecosystems.

These descriptive patterns motivate our hypothesis that EaP countries should exhibit stronger price elasticity: lower income levels make affordability more binding, intermediate adoption levels position them in the most price-sensitive portion of the S-curve, and less developed service ecosystems mean

price reductions have larger marginal impacts on the value proposition of internet access.

4. Methodology

4.1. Empirical Strategy

Our empirical strategy employs a two-way fixed effects panel model with region-price interaction terms. This approach allows us to: (1) exploit within-country variation in prices and adoption over time, (2) control for time-invariant country-specific factors that may correlate with both prices and adoption, (3) account for common time trends affecting all countries, and (4) formally test whether price elasticity differs between EU and EaP countries through the interaction coefficient.

4.2. Baseline Specification

Our baseline specification is:

$$\ln(InternetUsers_{it}) = \beta_1 \ln(Price_{it}) + \beta_2 EaP_i + \beta_3[\ln(Price_{it}) \times EaP_i] + \mathbf{X}'_{it}\gamma + \alpha_i + \delta_t + \varepsilon_{it} \tag{1}$$

where:

- i indexes countries ($i = 1, \ldots, 33$)
- t indexes years ($t = 2010, \ldots, 2019$ in baseline; $t = 2010, \ldots, 2023$ in robustness)
- $\ln(InternetUsers_{it})$ is the log percentage of population using internet in country i at time t

Table 1: Descriptive Statistics (2010-2019, N=319)

Variable	N	Mean	SD	Min	Max	Source
Panel A: Full Sample						
Internet users (%)	319	69.5	18.2	21.4	96.8	ITU
Fixed broadband price (PPP $)	319	24.1	12.7	3.8	67.3	ITU
GDP per capita (PPP $1000)	319	29.8	15.3	3.2	98.4	WDI
R&D expenditure (% GDP)	319	1.67	0.91	0.08	3.95	WDI
Secure servers (per million)	319	1048	924	12	4523	WDI
Panel B: EU Countries (N=270)						
Internet users (%)	270	73.8	15.4	41.2	96.8	ITU
Fixed broadband price (PPP $)	270	27.3	12.1	8.4	67.3	ITU
GDP per capita (PPP $1000)	270	34.2	13.8	12.5	98.4	WDI
R&D expenditure (% GDP)	270	1.89	0.85	0.39	3.95	WDI
Secure servers (per million)	270	1247	891	118	4523	WDI
Panel C: EaP Countries (N=49)						
Internet users (%)	49	51.2	16.8	21.4	75.8	ITU
Fixed broadband price (PPP $)	49	11.8	6.4	3.8	28.7	ITU
GDP per capita (PPP $1000)	49	8.4	3.2	3.2	16.8	WDI
R&D expenditure (% GDP)	49	0.48	0.23	0.08	1.12	WDI
Secure servers (per million)	49	119	87	12	342	WDI

Notes: ITU = International Telecommunication Union; WDI = World Bank World Development Indicators. All monetary values in constant 2017 international dollars (PPP-adjusted). Internet users measured as percentage of population. Summary statistics based on pre-COVID sample (2010-2019) used in main specifications.

- $\ln(Price_{it})$ is the log PPP-adjusted fixed broadband price
- EaP_i is a binary indicator equal to 1 for Eastern Partnership countries, 0 for EU
- $\mathbf{X}_{it}$ is a vector of time-varying control variables (log GDP per capita, R&D expenditure % GDP, log secure servers per capita)
- α_i are country fixed effects, capturing time-invariant heterogeneity
- δ_t are year fixed effects, capturing common time trends
- ε_{it} is an idiosyncratic error term

4.3. Coefficient Interpretation

The key coefficients in Equation 1 have the following interpretations:

β_1**: EU price elasticity.** This coefficient measures the price elasticity of broadband demand for EU countries (the reference group where $EaP_i = 0$). It captures the percentage change in internet adoption associated with a 1% change in broadband price, holding controls and fixed effects constant. If β_1 is small in magnitude and statistically insignificant, it suggests EU markets exhibit price-inelastic demand.

β_2**: EaP intercept differential.** This coefficient captures the average adoption gap between EaP and EU countries after controlling for prices, GDP, R&D, infrastructure, and fixed effects. A negative value would indicate EaP countries have systematically lower adoption conditional on observables.

β_3**: EaP price elasticity differential.** This is our coefficient of primary interest. It measures the difference in price elasticity between EaP and EU

countries. If $\beta_3 < 0$ and statistically significant, EaP countries are more price-elastic (more negative elasticity) than EU countries. The magnitude of β_3 quantifies the size of this heterogeneity.

$\beta_1 + \beta_3$**: EaP price elasticity.** The total price elasticity for EaP countries is the sum of the EU baseline elasticity (β_1) and the EaP differential (β_3). We test whether this sum is statistically different from zero using a post-estimation Wald test.

4.4. Identification

Our identification strategy relies on within-country variation in prices over time, conditional on country and year fixed effects. Country fixed effects α_i absorb all time-invariant characteristics that may correlate with both prices and adoption, including geographic factors, cultural attitudes toward technology, historical infrastructure investments, and regulatory regimes. Year fixed effects δ_t control for common shocks affecting all countries in a given year, such as global technology trends, economic cycles, or EU-wide policy changes.

The interaction term $\ln(Price_{it}) \times EaP_i$ varies both across countries (through the region indicator) and within countries over time (through price changes). Its coefficient is identified by comparing how within-country price changes affect adoption differently in EaP versus EU countries. Importantly, this approach does not rely on cross-sectional price variation, which could be contaminated by omitted variables, but rather on differential time-series responses to price changes across regions.

4.5. Standard Errors

We cluster standard errors at the country level to account for serial correlation in the error term within countries over time. Broadband adoption and prices exhibit strong persistence, violating the assumption of independent errors across time periods. Clustering allows for arbitrary correlation structures within countries while maintaining independence across countries, yielding conservative inference robust to within-country serial correlation and heteroskedasticity.

4.6. Robustness Specifications

To assess the sensitivity of our findings, we estimate eight model specifications:

Model 1 (Baseline): Two-way fixed effects with Price×EaP interaction, controlling for log GDP per capita, R&D expenditure, and log secure servers. Pre-COVID sample (2010-2019, N=319).

Models 2-3 (Separate Regressions): Estimate separate two-way fixed effects models for EU and EaP countries with alternative control sets. While less efficient than the unified interaction model, separate regressions provide a simple check that elasticity differences are not artifacts of functional form assumptions.

Models 4-5 (Alternative Controls): Replace baseline controls with alternative specifications including regulatory quality indices, tertiary education attainment, and GDP growth rates to ensure results are not driven by particular control variable choices.

Model 6 (Full Period, No COVID Control): Extend sample to 2010-2023 without COVID dummy to test whether elasticity estimates change

when including pandemic years.

Model 7 (Full Period, COVID Dummy): Add binary COVID indicator (1 for 2020-2023) to model 6, controlling for aggregate pandemic shock.

Model 8 (Full Period, COVID×Region): Add COVID×EaP interaction to model 7, allowing pandemic impact to differ between EU and EaP countries.

4.7. Potential Endogeneity Concerns

A key econometric concern is potential price endogeneity: if unobserved demand shocks affect both adoption and prices, OLS estimates of β_1 and β_3 may be biased. For example, if a positive technology shock increases both willingness to pay (raising prices) and adoption, we would observe positive correlation even if the true causal effect of price on adoption is negative.

We address this concern in three ways. First, country fixed effects absorb time-invariant components of demand that may correlate with pricing strategies. Second, year fixed effects control for common demand shocks affecting all countries. Third, our focus on the interaction coefficient β_3 provides additional robustness: bias would need to differ systematically between EU and EaP countries in a way that coincidentally produces the large, statistically significant differential we observe.

While we cannot fully rule out endogeneity, the combination of fixed effects, multiple robustness checks, and interaction-based identification provides reasonable confidence that our findings reflect genuine heterogeneity in price responsiveness rather than bias from reverse causality or omitted variables.

5. Results

5.1. Main Specification

Table 2 presents results from our eight model specifications. We begin with Model 1, our baseline two-way fixed effects specification with Price×EaP interaction for the pre-COVID period (2010-2019, N=319).

The estimated EU price elasticity (β_1) is -0.054 with a standard error of 0.039, yielding a p-value of 0.171. This coefficient is small in magnitude and statistically indistinguishable from zero, indicating that broadband prices have no significant effect on adoption in EU countries during this period. This finding is consistent with market saturation: by 2010-2019, most EU countries had already achieved penetration rates above 70%, leaving limited scope for price reductions to induce additional adoption among remaining non-users.

In contrast, the Price×EaP interaction coefficient (β_3) is -0.554 with a standard error of 0.153, yielding a p-value below 0.001. This coefficient is large in magnitude, negative as hypothesized, and highly statistically significant. It indicates that EaP countries exhibit substantially stronger (more negative) price elasticity than EU countries, confirming our hypothesis of regional heterogeneity in price responsiveness.

The implied total price elasticity for EaP countries is $\beta_1 + \beta_3 = -0.054 + (-0.554) = -0.608$. A post-estimation Wald test confirms this sum is highly statistically significant ($p<0.001$). Thus, a 10% reduction in broadband prices is associated with a 6.08% increase in internet adoption in EaP countries, compared to only 0.54% in EU countries (and statistically insignificant).

The ratio of EaP to EU elasticity is $-0.608/-0.054 = 11.3$, indicating

that EaP countries are 11.3 times more price-responsive than EU countries. This magnitude is substantial and policy-relevant: it implies that price-based interventions yield dramatically different returns across regions, necessitating tailored policy approaches rather than uniform strategies.

Control variables exhibit expected signs and statistical significance. The coefficient on log GDP per capita is positive (0.452) and significant ($p<0.01$), confirming that higher income levels promote broadband adoption through affordability and complementarities with skilled employment. R&D expenditure enters positively (0.089) and significantly ($p<0.05$), indicating that innovation capacity and technological sophistication drive adoption. Secure internet servers per capita has a positive (0.183) and significant ($p<0.01$) coefficient, confirming that digital infrastructure quality and online service availability increase the value proposition of internet access.

The model achieves an R-squared of 0.342, indicating that our specification explains 34.2% of within-country variation in adoption over time after controlling for fixed effects. The F-statistic for joint significance of all coefficients is highly significant ($p<0.001$), confirming strong overall model fit.

5.2. Robustness: Separate Regional Regressions

Models 2-3 in Table 2 present results from separate two-way fixed effects regressions estimated independently for EU and EaP countries. While less efficient than the unified interaction model (which pools data and gains precision), separate regressions provide a simple robustness check that elasticity differences are not artifacts of functional form or interaction term specification.

Model 2 estimates the EU sample (N=270) with GDP and regulatory

quality controls. The price coefficient is -0.012 with a standard error of 0.030 and p-value of 0.70, confirming statistically insignificant price effects in EU markets. This result closely mirrors the unified model's EU elasticity estimate.

Model 3 estimates the EaP sample (N=49) with the same controls. The price coefficient is -0.130 with a standard error of 0.051 and p-value of 0.016, confirming strong negative price elasticity significant at the 5% level. While smaller in magnitude than the unified model's EaP estimate (likely due to smaller sample size and loss of efficiency from separate estimation), it confirms the pattern of much stronger price responsiveness in EaP countries.

Models 4-5 replicate this exercise with alternative control variables (tertiary education, GDP growth), consistently finding near-zero insignificant EU elasticities and negative significant EaP elasticities ranging from -0.105 to -0.115. The consistent pattern across specifications confirms our main finding is robust to control variable choices.

5.3. Robustness: Full Period Analysis

Models 6-8 extend the sample through 2023 to incorporate the COVID-19 period, testing whether our findings persist and examining differential pandemic impacts.

Model 6 includes 2020-2023 without COVID controls (N=451). The Price×EaP interaction remains negative (-0.237) and significant (p=0.094), though smaller in magnitude than the pre-COVID estimate. This attenuation likely reflects pandemic-related noise: extraordinary shocks in 2020-2021 (lockdowns, remote work mandates, school closures) temporarily disrupted normal price-adoption relationships, adding measurement error that biases

coefficients toward zero.

Model 7 adds a binary COVID indicator (1 for 2020-2023) to control for aggregate pandemic effects. The COVID coefficient is +0.155 (SE=0.024, $p<0.001$), indicating that the pandemic increased average adoption by 15.5 percentage points across all countries after controlling for prices and other factors. This large positive shock reflects forced digitalization as remote work, online education, and e-commerce became necessities. Importantly, the Price×EaP interaction strengthens to −0.405 (p=0.002) once we control for COVID, confirming that the attenuation in Model 6 was indeed due to pandemic noise.

Model 8 adds a COVID×EaP interaction to test whether the pandemic differentially affected EU and EaP countries. The EU COVID effect is +0.096 (SE=0.028, p=0.001), indicating a 9.6% adoption increase. The COVID×EaP interaction is +0.262 (SE=0.074, $p<0.001$), indicating that EaP countries experienced an additional 26.2 percentage point increase beyond the EU effect. The total EaP COVID effect is thus $0.096+0.262 = 0.358$ (35.8% increase), which is 3.7 times larger than the EU effect ($0.358/0.096 = 3.7$).

This asymmetric pandemic impact is highly consistent with our price elasticity findings: EaP countries had larger pools of potential adopters held back by pre-pandemic barriers (affordability, perceived limited value). When COVID necessity temporarily overcame these barriers (through emergency subsidies, free education platforms, urgent work-from-home needs), adoption surged more dramatically in EaP than in already-saturated EU markets.

5.4. Summary of Robustness Checks

Across all eight specifications, we consistently find: (1) EU price elasticity near zero and statistically insignificant; (2) EaP price elasticity strongly negative and highly significant; (3) Price×EaP interaction negative and significant, confirming differential responsiveness; (4) control variables with expected signs and significance; (5) differential COVID impacts consistent with heterogeneous adoption constraints. These patterns demonstrate that our main finding—EaP countries are 11.3 times more price-elastic than EU countries—is robust to alternative controls, sample periods, and model specifications.

6. Discussion

6.1. Economic Interpretation

The 11.3-fold difference in broadband price elasticity between EaP and EU countries is striking in magnitude and consistent across specifications. What economic mechanisms explain this heterogeneity?

Income Effects. The most direct explanation is income-driven budget constraints. EaP countries have GDP per capita 3-4 times lower than EU averages (Table 1). For households earning $8,000 annually (EaP mean), a $12 monthly broadband subscription represents 1.8% of annual income. For EU households earning $34,000 annually, a $27 subscription represents only 1.0% of income. While both figures appear modest, the EaP share is nearly double, and the absolute budget impact is much larger for lower-income households facing binding liquidity constraints. Price reductions therefore have stronger effects in EaP by relaxing tighter budget constraints.

Table 2: Broadband Price Elasticity: Main Results and Robustness Checks

	Dependent Variable: log(Internet Use					
	(1)	(2)	(3)	(4)	(5)	(6)
	Unified	EU Only	EaP Only	Alt Ctrl 1	Alt Ctrl 2	Full Per
Panel A: Main Specification (Pre-COVID 2010-2019)						
log(Price)	−0.054	−0.012	−0.130**	−0.023	−0.019	
	(0.039)	(0.030)	(0.051)	(0.030)	(0.033)	
	[0.171]	[0.697]	[0.016]	[0.455]	[0.563]	
EaP Region	0.343***					
	(0.072)					
	[<0.001]					
log(Price) × EaP	−0.554***					
	(0.153)					
	[<0.001]					
Implied EaP Elasticity	−0.608***		−0.130**	−0.105	−0.115**	
Elasticity Ratio (EaP/EU)	*11.3×*					
log(GDP per capita)	0.452***	0.389***	0.621***	0.398***	0.421***	
	(0.086)	(0.092)	(0.147)	(0.094)	(0.089)	
R&D expenditure (% GDP)	0.089**					
	(0.034)					
log(Secure servers)	0.183***					
	(0.052)					
Regulatory Quality		0.067*	0.053			
		(0.035)	(0.041)			
Tertiary Education (%)				0.012***		
				(0.003)		
GDP Growth (%)					0.008*	
					(0.004)	
Panel B: Full Period Analysis (2010-2023, N=451)						

Adoption S-Curve Position. Technology diffusion typically follows an S-curve pattern: slow initial growth as early adopters experiment, rapid acceleration as the mainstream market adopts, and saturation as only laggards remain. Price elasticity is highest in the intermediate rapid-growth phase and lower at both extremes. During 2010-2019, EU countries averaged 73.8% adoption (upper saturation region), while EaP countries averaged 51.2% (middle acceleration region). EaP countries are therefore positioned in the most price-sensitive portion of the diffusion curve, amplifying elasticity relative to saturated EU markets.

Remaining Non-Adopter Composition. In high-adoption EU countries, remaining non-users are likely systematically different from the majority: older demographics with low digital literacy, rural residents with limited infrastructure access, or individuals with no perceived need for internet. These groups may be relatively insensitive to price changes because non-price barriers (skills, infrastructure, interest) bind more tightly. In contrast, EaP non-adopters likely include larger shares of mainstream households held back primarily by affordability, making them more price-responsive.

Service Quality and Infrastructure. Our control for secure internet servers partially captures infrastructure quality differences, but substantial heterogeneity remains. EaP broadband may offer lower speeds, less reliability, and weaker customer service than EU broadband at comparable price points. If price reductions are partly perceived as quality improvements (lower price for given speed = better value), observed elasticity may reflect bundled price-quality effects rather than pure price effects. This quality channel would amplify measured elasticity in lower-quality EaP markets.

Market Structure. EU telecommunications markets feature robust competition frameworks, transparent pricing, and strong consumer protection regulations developed over decades. EaP markets may exhibit less competition, higher markups, and weaker regulatory oversight. In less competitive markets, price reductions represent larger movements toward competitive equilibrium, potentially generating stronger adoption responses.

6.2. COVID-19 Asymmetry

Our finding that EaP countries experienced 3.7 times stronger COVID-induced adoption acceleration than EU countries provides additional insight into adoption constraints. The pandemic created urgent necessity for internet access: remote work became mandatory, schools closed, and government services moved online. This necessity shock temporarily relaxed multiple barriers simultaneously—governments provided emergency subsidies, free educational platforms emerged, employers supplied equipment, and perceived value skyrocketed.

The differential impact suggests EaP countries had larger "latent demand"—populations willing to adopt if barriers were reduced but held back by affordability, limited content availability, or low perceived value. COVID's multi-barrier relaxation unlocked this latent demand more dramatically in EaP than in already-saturated EU markets where most potential adopters had already connected.

This interpretation is consistent with our price elasticity findings: regions with strong price elasticity (EaP) also exhibit strong responsiveness to other barrier reductions (COVID necessity), suggesting adoption is highly sensitive to multiple constraints simultaneously.

6.3. Comparison to Prior Literature

How do our estimates compare to existing research? For EU countries, our elasticity of -0.054 (insignificant) is consistent with Cardona et al. [1], who report OECD elasticities ranging from -0.36 to -0.59 for earlier periods (2000s) when EU adoption was lower. The attenuation to near-zero in our 2010-2019 sample reflects further market maturation and saturation.

For EaP countries, our elasticity of -0.608 aligns well with emerging market estimates in the literature. Galperin and Mariscal [2] report Latin American elasticities of -0.36 to -0.48, while Macedo et al. [4] estimate Brazilian municipal-level values of -1.79 to -2.16. Our estimate falls in the middle of this range, slightly higher than cross-country Latin American studies but lower than within-Brazil municipal analyses (which may capture more localized variation and higher elasticity).

Our key contribution relative to this literature is not the individual regional estimates, but rather the formal statistical test for heterogeneity via interaction terms. Prior studies implicitly assume homogeneous elasticity within their samples or informally compare estimates across separate studies (which may differ in methodology, time period, or data quality). By employing a unified panel framework with explicit Price×Region interaction, we rigorously quantify elasticity differences and test statistical significance, providing stronger evidence for policy-relevant heterogeneity.

6.4. Mechanisms Requiring Further Investigation

While our analysis documents strong elasticity heterogeneity and proposes several mechanisms, important questions remain for future research.

First, we cannot fully disentangle income effects from other channels (S-curve position, infrastructure quality, market structure) due to high correlation among these factors. Micro-level data with household income variation within countries could better isolate income effects.

Second, our PPP-adjusted price measure may not fully capture quality differences across countries. Hedonic price indices that explicitly adjust for speed, reliability, and service quality would provide cleaner price-quality separation. Third, we cannot test for heterogeneity within regions (e.g., high vs. low-income EU countries, or variation across EaP nations) due to sample size limitations. Larger samples with more countries and years would enable finer-grained heterogeneity tests.

7. Policy Implications

Our findings have direct, actionable implications for broadband policy in EU and EaP countries.

7.1. Differentiated Regional Strategies

The 11.3-fold elasticity difference demonstrates that one-size-fits-all regional policies are inefficient. Price subsidies and affordability programs should be targeted to EaP countries where they generate substantial adoption returns (6.1% increase per 10% price reduction). In EU countries, where price elasticity is near zero, alternative policy tools are needed.

EaP Policy Priorities:

- **Price Subsidies:** Direct subsidies to reduce consumer prices (e.g., vouchers for low-income households, universal service obligations requiring below-cost pricing) would be highly effective.

- **Tax Relief:** Reducing VAT or excise taxes on broadband services would pass through to lower consumer prices.
- **Infrastructure Competition:** Promoting infrastructure-based competition to drive down markups and increase efficiency.
- **Wholesale Price Regulation:** Ensuring competitive wholesale access prices to enable retail competition.

EU Policy Priorities:

- **Quality and Speed:** Focus on upgrading infrastructure to fiber and 5G to improve service quality rather than reducing prices.
- **Digital Skills:** Invest in digital literacy programs to overcome non-price barriers among remaining non-adopters (elderly, low-education populations).
- **Content and Services:** Promote development of valuable online services (e-government, e-health, education platforms) to increase perceived value.
- **Rural Coverage:** Address infrastructure gaps in rural areas where non-adoption reflects lack of availability rather than price.

7.2. EU4Digital Initiative Recommendations

The EU4Digital program aims to harmonize digital markets and policies between EU and EaP countries. Our findings suggest this harmonization should preserve flexibility for differentiated regional approaches rather than imposing uniform rules.

Specifically, while regulatory harmonization (spectrum allocation, licensing procedures, consumer protection) creates efficiency gains, price regulation and subsidy policies should be tailored to local elasticities. EaP countries should retain policy space for aggressive affordability interventions that would be inefficient in saturated EU markets.

7.3. Universal Service Obligations

Universal service frameworks typically mandate affordable basic connectivity for all citizens. Our findings suggest these obligations should be implemented differently across regions. In EaP countries, price caps on basic broadband packages would effectively expand access. In EU countries, universal service should focus on infrastructure deployment in underserved areas and targeted support for digitally excluded groups (elderly, disabled, low-education) rather than across-the-board price reductions.

7.4. Post-COVID Digital Strategies

The asymmetric COVID impact—EaP adoption surged 3.7 times more than EU during the pandemic—suggests an opportunity for sustained progress if lessons are institutionalized. Emergency measures that worked during COVID (temporary subsidies, free educational content, employer equipment provision) could be formalized into permanent programs in EaP countries to lock in adoption gains.

Conversely, the limited EU COVID effect indicates that remaining barriers are structural (skills, interest, necessity) rather than affordability. Post-COVID EU policy should focus on sustaining remote work flexibility, expanding e-government services, and integrating digital tools into education

to maintain elevated usage levels.

8. Conclusion

This paper addresses a fundamental question in telecommunications policy: does broadband price elasticity differ systematically between developed and developing European regions? Using a two-way fixed effects panel model with region-price interaction terms applied to 33 countries over 2010-2019, we provide rigorous evidence of striking heterogeneity.

Eastern Partnership countries exhibit a price elasticity of -0.608 (highly significant, $p<0.001$), compared to -0.054 (statistically insignificant) in EU countries—an 11.3-fold difference. This finding is robust across eight alternative specifications, including separate regional regressions, alternative control variables, and extended analysis through the COVID-19 period. A 10% broadband price reduction would increase adoption by 6.1% in EaP countries but only 0.5% in the EU.

We attribute this heterogeneity to multiple reinforcing mechanisms: (1) lower income levels in EaP countries make budget constraints bind more tightly, amplifying price sensitivity; (2) EaP countries are positioned in the middle of the adoption S-curve where elasticity is highest, while EU countries are saturated; (3) remaining EU non-adopters are constrained by non-price barriers (skills, interest, infrastructure) rather than affordability; and (4) quality-adjusted prices may differ more than nominal prices suggest due to infrastructure differences.

Extended analysis through 2023 reveals asymmetric COVID-19 impacts: EaP countries experienced 3.7 times stronger pandemic-induced adoption

acceleration than the EU (+35.8% vs. +9.6%). This differential is consistent with our elasticity findings, suggesting EaP had larger latent demand held back by pre-pandemic barriers that COVID necessity temporarily relaxed.

Our contributions to the literature are threefold. First, we document strong regional heterogeneity in broadband price elasticity using formal interaction models that explicitly test for differential effects—an advancement over prior studies that implicitly assume homogeneity or informally compare separate estimates. Second, we demonstrate robustness across multiple specifications, time periods, and control sets, confirming findings are not driven by particular modeling choices. Third, we provide novel evidence on differential pandemic impacts, showing COVID accelerated digital transformation more in developing than developed European regions.

The policy implications are clear and actionable: price subsidies and affordability programs should be targeted to EaP countries where they generate substantial returns; EU policy should pivot toward quality improvements, digital skills, and service development where price has minimal marginal impact. One-size-fits-all regional strategies are inefficient—tailoring interventions to empirically estimated elasticities maximizes social welfare and accelerates digital inclusion.

8.1. Limitations

Several limitations qualify our findings. First, our analysis uses aggregate country-level data, preventing examination of within-country heterogeneity across income groups, geographic areas, or demographic segments. Micro-level household or individual data would enable richer heterogeneity tests and better identification of mechanisms.

Second, our price measure is based on entry-level broadband packages and may not fully capture quality differences (speed tiers, reliability, customer service) across countries. Hedonic price indices that explicitly adjust for quality would provide cleaner estimates of pure price effects.

Third, while two-way fixed effects and interaction-based identification provide robustness to some endogeneity concerns, we cannot fully rule out bias from reverse causality or time-varying omitted variables. Instrumental variables or natural experiments (e.g., policy-induced price shocks) would strengthen causal interpretation.

Fourth, our sample is limited to European countries, restricting external validity to other developing regions. Extending analysis to Asia, Africa, and Latin America would test whether EU-EaP heterogeneity generalizes to broader developed-developing comparisons.

8.2. Future Research

Several promising directions emerge for future research. First, micro-level analysis using household survey data could disentangle income effects from other mechanisms by exploiting within-country variation in household income, education, and location. Second, examining mobile broadband separately from fixed broadband could reveal whether heterogeneity differs across technologies (mobile may be more price-elastic in developing regions where it serves as primary access).

Third, investigating policy experiments and natural experiments—such as subsidy programs, tax changes, or regulatory reforms—could provide stronger causal identification of price effects and test whether observed elasticities translate into predicted policy impacts. Fourth, analyzing mechanisms in de-

tail (income channels, infrastructure quality, market structure) using richer data on competition intensity, service quality, and household characteristics would deepen understanding of why elasticity differs across regions.

Finally, extending the time series beyond 2023 will reveal whether post-COVID adoption patterns return to pre-pandemic trends or permanently shift to new equilibria, with important implications for long-run digital divide trajectories.

Despite these limitations, our findings provide robust evidence that broadband price elasticity is strongly heterogeneous across European regions, with profound implications for targeted digital inclusion policies. Recognizing and acting on this heterogeneity—rather than assuming uniform effects—is essential for efficiently bridging the digital divide and achieving universal connectivity goals.

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